

Ruichen Xu

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 [LinkedIn](#) |  [ru1ch3n](#) |  [Google Scholar](#) | [OpenReview](#)

SUMMARY

Ph.D. candidate in Applied Mathematics & Statistics at Stony Brook University. Research interests include **LLMs/agents for AI4Science**, **generative AI** for physics- and constraint-aware sampling, and **neural operators** for partial observations and inverse problems. I build reproducible pipelines and conduct large-scale experiments on SLURM/CUDA HPC clusters.

EDUCATION

- **Stony Brook University (The State University of New York)** Aug 2022 – Present (Expected 2027)
Ph.D., Computational Applied Mathematics (GPA 4.00/4.00) Stony Brook, NY, USA
- **Courant Institute of Mathematical Sciences, New York University (NYU)** Sep 2020 – May 2022
M.S., Mathematics (GPA 4.00/4.00) New York, NY, USA
- **University of California, Davis** Sep 2019 – Jun 2020
M.S., Statistics (GPA 3.94/4.00) Davis, CA, USA
- **Beijing University of Chemical Technology (BUCT)** Sep 2015 – Jun 2019
Bachelor, Financial Mathematics (89.33/100); Minor: Commercial Management Beijing, China

PROFESSIONAL EXPERIENCE

- **AI3, Stony Brook University** Jun 2026 – Jul 2026
REU Research Mentor Stony Brook, NY, USA
 - Mentors REU students on AI4Science research projects at AI3, including project scoping, literature review, experiment design, and technical implementation.
 - Guides students on reproducible research workflows, code organization, experiment tracking, and presentation preparation.

PROJECTS

- **Joint-Embedding Predictive Learning for Scientific Data** Jan 2026 – Present
Tools: Python, PyTorch; Topics: JEPA, neural operators, graph learning, AI4Science
 - Develops JENO for sparse-observation PDE inference and HP-JEPA for multi-resolution graph representation learning.
 - Builds reproducible student-teacher/EMA training, ablation, and evaluation pipelines on GPU clusters.
- **PartialObs-PDEBench & AI4Physics** 2025 – Present
Tools: Python; Topics: partial observations, benchmarking, diffusion, neural operators 
 - Maintains a curated AI4PDE/AI4SDE collection and a benchmark for sparse, irregular, sensor-style, and masked observations.
 - Provides unified datasets, configurations, and metrics for neural operators, diffusion models, and physics-aware baselines.
- **ORACLE: LLM-Guided Molecular Optimization for Structure-Based Drug Design** 2025 – Present
Tools: Python, PyTorch; Topics: molecular editing, simulated annealing, multi-objective optimization
 - Combines LLM-proposed molecular edits with simulated annealing for validity-constrained, multi-objective optimization.
 - Evaluates docking, QED, SA-score, diversity, and Pareto trade-offs through reproducible ablations.

PUBLICATIONS

Conferences

- [c.1] **HP-JEPA: Hierarchical Partitioning for Multi-Resolution Graph Joint-Embedding Predictive Learning.** Ruichen Xu, Jingxiang Qu, Wenhan Gao, Jiaxing Zhang, Linsey Pang, Ravid Shwartz-Ziv, Yann LeCun, Yuefan Deng. *In submission to KDD 2027.*
- [c.2] **JENO: Full-Field Latent Prediction for Sparse Inverse PDE Inference.** Ruichen Xu, Siyao Wang, Fang Wan, Jingxiang Qu, Tian Xie, Jiaxing Zhang, Linsey Pang, Wenhan Gao, Ravid Shwartz-Ziv, Yuefan Deng. *In submission to NeurIPS 2026.*
- [c.3] **IAGA: Identity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** Jingxiang Qu, Wenhan Gao, Ruichen Xu, Yi Liu. *ICLR 2026 (Accepted).* ([OpenReview](#))

Conferences (continued)

- [c.4] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** Ruichen Xu, Claire Yu, Zongyu Wu, Siyao Wang, Luoyao Chen, Georgios Kementzidis, Haochun Wang, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#))
- [c.5] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** Wenhan Gao, Ruichen Xu, Yuefan Deng, Yi Liu. *ICLR 2025*. ([OpenReview](#)) ([ICLR Page](#))
- [c.6] **Kolmogorov–Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, R. Xu[†], L. Chen, G. Kementzidis, S. Wang, Y. Deng. *IJCNN 2025*. ([IEEE](#)); [†]co-first author; project lead.
- [c.7] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** Maximilian Cederholm, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).
- [c.8] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** Albert Ding, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).
- [c.9] **APOD: Adaptive PDE-observation diffusion for physics-constrained sampling.** Ruichen Xu, Haochun Wang, Georgios Kementzidis, Chenhao Si, Yuefan Deng. *ICML 2025 Workshop on Assessing World Models (Poster)*. ([OpenReview](#))
- [c.10] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** Ruichen Xu, Kai Li, Haochun Wang, Georgios Kementzidis, Wei Zhu, Yuefan Deng. *2nd AI for Math Workshop @ ICML 2025 (Poster)*. ([OpenReview](#))
- [c.11] **An Iterative Framework for Generative Backmapping of Coarse-Grained Proteins.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *ICML 2025 GenBio Workshop (Poster)*. ([OpenReview](#))

Journals

- [j.1] **Multistep Generative Backmapping of Coarse-Grained Structures.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *Computer Physics Communications*, **327**, 110286 (2026). ([DOI](#)) ([Elsevier Share Link](#))
- [j.2] **Dynamic Schwartz–Fourier Neural Operator for Enhanced Expressive Power.** Wenhan Gao, Jian Luo, Ruichen Xu, Yi Liu. *TMLR 2025*. ([OpenReview](#))
- [j.3] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** Wenhan Gao, Ruichen Xu, Hong Wang, Yi Liu. *TMLR 2024*. ([OpenReview](#))

TEACHING EXPERIENCE

- **Stony Brook University – Department of Applied Mathematics & Statistics (AMS)** *Aug 2022 – Present*
Teaching Assistant & Instructor Stony Brook, NY, USA
 - **Teaching Assistant (Aug 2022 – May 2025):** AMS 310 (F'22, S'23), AMS 326 (S'24), AMS 510 (F'24; led recitations), AMS 595 (F'24), AMS 528 (S'25), AMS 502 (S'25); ran discussions, grading, and office hours.
 - **Instructor (Jan 2023 – Present):** AMS 394 (W'23, W'24, Su'24, Su'25), AMS 326 (Su'25), AMS 361 (W'26); designed/delivered lectures, created and graded assessments, and mentored students.
- **Courant Institute of Mathematical Sciences, New York University (NYU)** *Fall 2020 – May 2022*
Recitation Leader & Grader New York, NY, USA
 - **Recitation Leader (Jan 2021 – May 2022):** Mathematics for Economics II (Sp'21, Sp'22); Probability, Statistics & Decision Making (Fa'21, Sp'22). Led recitations, held office hours, and graded.
 - **Grader (Fall 2020, Fall 2021):** Analysis (Fa'20); Special Topics (Fa'21).

TALKS & PRESENTATIONS

- **Lightning Talk – Diffusion4PDE**, IACS Annual Meeting, 2025; **Lightning Talk – Active Learning for Neural Operators**, IACS Annual Meeting, 2024; **Student Seminar Speaker – AI for PDEs**, IACS Student Seminar, 2025.

ACADEMIC SERVICE

- Reviewer: *IJCNN 2025/2026; TNNLS 2025; ICLR 2026; Gold Reviewer, ICML 2026; Reviewer, NeurIPS (formerly NIPS) 2026; Neurocomputing 2026.*